

Group02 - PA2 - User Research

Project Information

- **Course:** Human - Computer Interaction
- **Topic:** Running Music Coach - Adaptive Music and Eyes-Free Interaction for Runners
- **Team:** Group02
- **Lecturer:** Le Khanh Duy
- **Date:** July 2026

Members

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1. Research Objectives

The primary objective of this research is to understand how recreational runners use music, mobile applications, and wearable devices during running, and to identify pain points related to pace monitoring, audio interruptions, and eyes-free interaction.

Specifically, this research aims to:

- understand how runners currently use running applications and music services during a typical session;
- investigate how music influences motivation, rhythm, and unintentional pace changes;
- explore how runners monitor pace, cadence, and effort without disrupting their running flow;
- identify difficulties when interacting with phones or smartwatches while moving;
- document the workarounds runners already use to reduce friction and maintain focus.

2. Research Scope and Stakeholders

2.1 Scope

This research focuses on **young recreational runners aged 18-25**, primarily **university students**, who run at least twice per week, listen to music while running, and use a smartphone and/or smartwatch with at least one running application. Sessions typically take place outdoors in parks, residential routes, or fixed neighborhood loops.

2.2 Stakeholders

Primary stakeholders

- Recreational runners
- Phone-first runners
- Watch-assisted runners
- Solo runners who rely on music for motivation

Secondary stakeholders

- Running app providers such as Strava or Garmin
- Music services such as Spotify
- Wearable manufacturers such as Apple Watch or Huawei

3. Research Techniques and Process

3.1 Research Methods Used

This study used two complementary user research methods:

1. Semi-structured interviews
2. Contextual observation

These methods were chosen because they capture both what users say and what they actually do while running or when recalling real running situations.

3.2 Semi-Structured Interviews

- **Participants:** 7 participants (P01-P07)
- **Duration:** approximately 20-30 minutes each
- **Recording approach:** note-taking using structured interview summary templates with participant IDs only
- **Topics covered:**
 - how users monitor running performance;
 - how they interact with phone or smartwatch during running;
 - how music affects motivation and pace;
 - what interruptions break running flow;
 - what workarounds they use.

The interviews were semi-structured so the team could ask the same core questions across participants while still probing for unexpected stories, frustrations, and coping strategies.

3.3 Observation

Observation was conducted using a structured checklist during or around running sessions. Notes were recorded with participant IDs only and later compared with interview responses. The team focused on:

- how often users looked at phone or smartwatch;
- when they paused, resumed, or changed tracks;
- whether sweat, vibration, and movement affected touch accuracy;
- how users reacted to voice coach, alerts, and calls;
- what safety or convenience workarounds they used.

3.4 Participant Summary

ID	Profile	Experience	Devices / Apps	Key insight
P01	IT student, year 3	6 months, 2-3x/week	Strava, phone only	Screen checks cause dizziness; wants proactive reminders
P02	Network student, year 4	6 months, 3x/week	Spotify, Garmin, watch	Watch reduces phone use; selective pace checking
P03	Student, year 2	~1 year, 3x/week	Strava, Spotify, phone + earbuds	Voice alerts break music; sweaty touch is major friction
P04	Tech fresher, age 22	3-4 months, ~3x/week	Apple Watch, phone music	Wants subtle haptic/audio pace cues instead of voice
P05	Student, year 3	~6 months, 3-4x/week	Huawei watch, Strava	Disables voice coach; distrusts sync inaccuracies
P06	Student, year 3	< 6 months, 5-6x/week	Strava, Huawei watch	Wants hands-free "agent-like" support
P07	UEH student, year 3	> 1.5 years irregular	Strava, Health, phone + earbuds	Carries phone due to pocket vibration; music drives motivation

4. Research Findings

Analysis of interview and observation data revealed four recurring themes.

4.1 Music Is a Major Motivational Tool

- Music is a major motivational tool, especially for solo runners.
- When music is interrupted by voice coach, notifications, or system audio ducking, runners often lose mood and rhythm.
- Some runners reported that without music they feel bored faster or may end the session early.

4.2 Visual Pace Checking Is Disruptive

- Runners want to know whether they are running too fast, too slow, or overexerted.
- However, looking at a phone or watch during movement is disruptive and sometimes unsafe.
- Participants reported dizziness, loss of breathing rhythm, and reduced environmental awareness when checking screens mid-run.

4.3 Device Interaction Is a Repeated Friction Point

- Sweaty hands make unlock, pause/resume, and swipe gestures difficult.
- Small smartwatch displays increase the chance of mis-taps.
- Touch-based interaction is unreliable in a high-motion, low-attention context.

4.4 Users Already Create Workarounds

Common workarounds include:

- preparing playlists in advance;
- enabling Do Not Disturb / Focus mode;
- disabling voice coach;
- relying on smartwatches instead of phones;
- checking pace only at selected moments.

These workarounds help, but they only partially solve the problem.

5. Personas

Persona 1: Anh - Phone-First Runner

- **Age:** 20
- **Occupation:** University student
- **Running experience:** ~1 year, 3 sessions/week
- **Primary tools:** Strava, Spotify, phone + Bluetooth earbuds

Background: Anh runs to stay healthy and relieve stress after classes. He depends on his phone for both tracking and music and does not own a smartwatch.

Goals

- maintain a stable running pace without constantly taking out his phone;
- keep a seamless music experience throughout the run;
- know when he is pushing too hard without breaking flow.

Frustrations

- sweaty hands make unlock and pause/resume difficult;
- notifications and voice prompts interrupt music;
- looking at the screen while running causes distraction or dizziness.

Persona 2: Binh - Watch-Assisted Runner

- **Age:** 22
- **Occupation:** Technology fresher
- **Running experience:** 3-4 months, ~3 sessions/week
- **Primary tools:** Apple Watch, phone for music, Bluetooth earbuds

Background: Binh uses a smartwatch to reduce phone dependency while running. He checks pace or heart rate from his wrist and keeps the phone in his pocket.

Goals

- track pace, heart rate, and distance conveniently;
- stay motivated through music;
- avoid awkward watch interactions and unwanted interruptions.

Frustrations

- small watch screen and wet fingers make taps unreliable;
- alerts duck music volume and break emotion;
- fast music can unintentionally increase pace and cause breathlessness.

Persona 3: Minh - Music-Driven Solo Runner

- **Age:** 21
- **Occupation:** University student
- **Running experience:** 6-12 months, 3-5 sessions/week
- **Primary tools:** Strava or native running app, Spotify, earbuds or watch music

Background: Minh often runs alone and treats music as the main source of motivation. He prefers to stay in flow and only wants feedback when it actually matters.

Goals

- stay motivated throughout solo runs;
- maintain an even pace and finish the session without quitting early;
- focus on body and music rather than device management.

Frustrations

- losing music or having it interrupted quickly reduces motivation;
- fast songs can pull pace upward unintentionally;
- voice coach and mistimed alerts destroy flow.

6. Task Analysis and Work Models

6.1 Hierarchical Task Analysis (HTA)

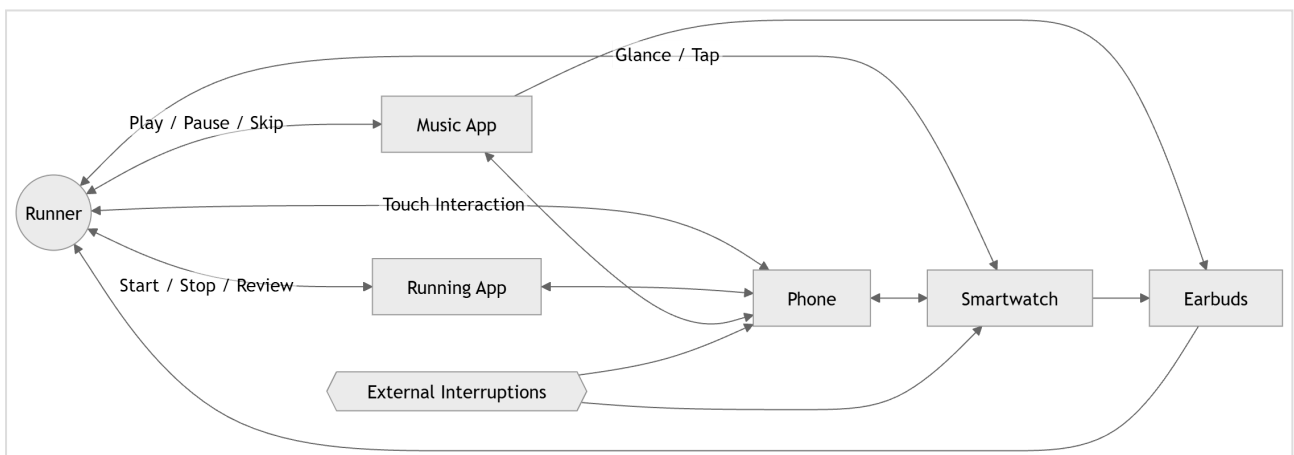
1. Complete a recreational running session
 - 1.1 Prepare for the run
 - 1.1.1 Choose route and goal
 - 1.1.2 Open running app and music app
 - 1.1.3 Select playlist or audio source
 - 1.1.4 Attach phone / wear watch / connect earbuds
 - 1.2 Start the session
 - 1.2.1 Start activity tracking

- 1.2.2 Start music playback
- 1.2.3 Begin warm-up / running movement
- 1.3 Run while monitoring state
 - 1.3.1 Maintain pace and breathing rhythm
 - 1.3.2 Respond to music tempo and motivation
 - 1.3.3 Notice fatigue, thirst, or environmental conditions
- 1.4 Check performance or adjust audio (as needed)
 - 1.4.1 Glance at phone or watch
 - 1.4.2 Unlock screen or navigate UI
 - 1.4.3 Pause, resume, skip track, or change volume
 - 1.4.4 React to voice coach / notification / call
- 1.5 Recover from interruption or overexertion
 - 1.5.1 Slow down, walk, or stop briefly
 - 1.5.2 Re-establish breathing and rhythm
 - 1.5.3 Decide whether to continue or end early
- 1.6 End the session
 - 1.6.1 Stop tracking
 - 1.6.2 Review summary stats
 - 1.6.3 Save or sync activity data

The most problematic low-level tasks are 1.4.1-1.4.4, where attention shifts away from body and environment toward device UI or intrusive audio.

6.2 Flow Model

Figure 1. Current flow model of a recreational running session



Main actors

- Runner
- Running application
- Music application
- Phone
- Smartwatch (optional)
- Earbuds
- External interruptions such as notifications, calls, GPS issues, and weather

Current flow

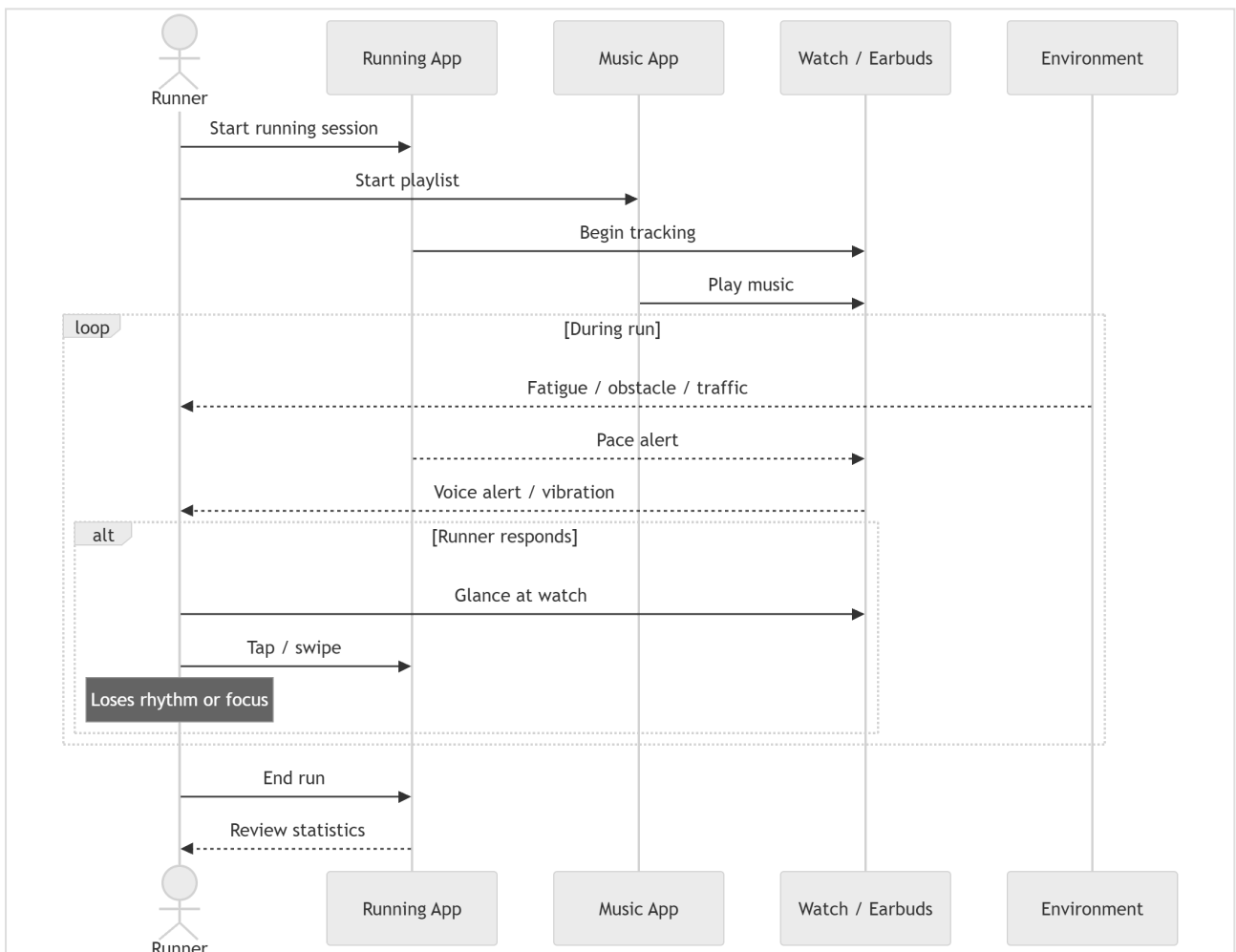
1. Runner prepares apps, music, and devices before leaving.
2. Runner starts tracking and music almost simultaneously.
3. During the run, the runner mostly maintains pace by feel and music rhythm.
4. When uncertain or interrupted, the runner checks or controls phone/watch.
5. Audio cues, messages, or mis-taps may force the runner to slow down or stop.
6. At the end, the runner stops tracking and reviews/syncs data.

Problems in current flow

- motivation and pace feedback are split across separate apps and devices;
- visual checks and audio interruptions break the running loop;
- workaround settings are manual and inconsistent;
- watch users still depend on phone for some music/control tasks.

6.3 Sequence Model

Figure 2. Current sequence model of running with music and pace monitoring



Current sequence

1. Decide to run
2. Open running app
3. Open music app / select playlist

4. Wear earbuds and start tracking
5. Run while listening to music
6. Notice pace, fatigue, or interruption
7. Glance at phone/watch or hear voice alert
8. Unlock / tap / swipe to respond
9. Lose rhythm or mood
10. Recover pace or end session early
11. Stop activity and review stats

Breakdowns

- steps 7-9 often happen at the worst moment, when the runner is already in flow;
- voice coach and notification ducking create unplanned audio breaks;
- wet hands and vibration make response actions slow and error-prone.

6.4 Artifact Model

Current artifacts

- smartphone;
- smartwatch (optional);
- Bluetooth earbuds;
- running app UI;
- music playlist / streaming app;
- lock screen, notification banners, voice coach audio;
- waist pouch, armband, or hand-carry phone;
- Do Not Disturb / Focus mode settings.

Problems with current artifacts

- phone and watch UIs assume stable touch and visual attention;
- voice coach treats feedback and music as separate channels;
- playlists are static and may not match target cadence or effort;
- notifications are not context-aware for active running;
- syncing between watch and apps can reduce trust in the data.

7. Representative Quotes

- P01: "Looking at the phone while running makes me dizzy and reduces performance."
- P03: "The music volume suddenly drops, then the app voice announces pace and distance."
- P04: "I wish pace reminders were a gentle vibration or subtle sound blended into music."
- P06: "I want an intelligent agent to do tasks that are hard to perform quickly while running."
- P07: "If I look down to adjust the phone while running, I lose breathing rhythm and focus."

8. Research Implications

The design opportunity is not simply to add more features. The stronger direction is to support running flow through:

- adaptive music behavior;
- low-attention, eyes-free pace feedback;
- reduced mid-run interaction burden.

9. Limitations

- The participant pool is homogeneous in age and student status.
- Findings are most representative of young recreational runners.
- Broader validation is needed for older users, non-students, and different running contexts.